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Operating Systems for Intelligent Systems (ENGS2306)

Lecture 4: Linux Users and Groups Management

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Lecture 4: Linux Users and Groups Management

Today's Objectives:

1. Understand the different types of users and their ID ranges (UIDs/GIDs).
 2. Learn to create, modify, and delete user accounts.
 3. Learn to create and manage groups to control permissions efficiently.
 4. Apply these concepts to secure intelligent systems (e.g., IoT devices).
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Recap: From Permissions to People

- In **Lecture 3**, we learned about file permissions (rwx) and how to change them (chmod) and their ownership (chown).
 - We asked the questions: **Who** can do **what** to a file?
 - Today, we focus on the "**Who**". We will learn how to create and manage the user accounts and groups that own files and are granted permissions.
 - Effective user management is the foundation of a secure and organized Linux system, especially for multi-purpose devices like IoT gateways or shared development environments.
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Understanding Users and UIDs

In Linux, a "user" is not just a name. It is an entity with specific privileges, identified by a number called a **User ID (UID)**.

- The system uses UIDs, not usernames, for tracking ownership and permissions.
- Every user also has a primary **Group ID (GID)**.

Standard UID Ranges in Ubuntu:

- **0 (root):** The **superuser**. This user has unlimited access and can do anything on the system. Use its power with caution, typically via the sudo command.
- **1-999 (System Users):** Reserved for services and applications running in the background (daemons). For example, a web server might run as the user www-data. These users usually cannot log in.
- **≥1000 (Regular Users):** For normal human users who log in to the system. You are a regular user.

Creating and Deleting Users

You must use `sudo` for these commands as they modify core system files.

1. Create a User: `useradd`

This command creates a new user account.

- **Syntax:** `sudo useradd [options] <username>`
- **Important Option:** `-m` creates a home directory for the user at `/home/<username>`.
- **Example:**

```
# Create a user named 'tariq' with a home directory
sudo useradd -m tariq
```

2. Delete a User: `userdel`

This command removes a user account.

- **Syntax:** `sudo userdel [options] <username>`
- **Important Option:** `-r` removes the user's home directory and mail spool.
- **Example:**

```
# Remove the user 'tariq' and his home directory
sudo userdel -r tariq
```

Managing User Passwords & Accounts

A new user account is locked until a password is set.

1. Set/Change a Password: `passwd`

- Root can set a password for any user:

```
# Set the password for the new user 'tariq'
sudo passwd tariq
# You will be prompted to enter and confirm the new password
```
- A regular user can change their own password:

```
# If logged in as 'tariq', this changes tariq's password
passwd
```

2. Modify a User: `usermod`

Used to change account details. For example, you can lock an account to temporarily disable it without deleting it.

- **Lock an account:** `sudo usermod -L <username>` (L for Lock)
 - **Unlock an account:** `sudo usermod -U <username>` (U for Unlock)
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Introduction to Groups

A **group** is a collection of users. Groups simplify permission management.

Why use groups?

Imagine you have a project directory and 5 developers who need access.

- **Without groups:** You would have to grant permissions to each of the 5 users individually. If a new developer joins, you have to repeat the process.
- **With groups:** You create one group (e.g., `developers`), add all 5 users to it, and grant permissions to the *group*. When a new developer joins, you just add them to the group, and they automatically get the correct permissions.

In Intelligent Systems: You can create a group called `sensor_apps` and give it access to specific hardware interfaces, ensuring only approved applications can read sensor data.

Managing Groups

1. Create a Group: `groupadd`

- **Syntax:** `sudo groupadd <groupname>`
- **Example:**

```
sudo groupadd iot_team
```

2. Add a User to a Group: `usermod`

This is the most common group management task.

- **Syntax:** `sudo usermod -aG <groupname> <username>`
- `-a` means **append**. It adds the user to the new group without removing them from other groups. **This is very important!**
- `-G` specifies the secondary group(s).
- **Example:**

```
# Add user 'tariq' to the 'iot_team' group
sudo usermod -aG iot_team tariq
```

3. Verify Group Membership: Use the `id` or `groups` command.

```
id tariq
groups tariq
```

Practical Scenario: Setting Up a Project Team

Follow these steps to create a secure environment for a new project.

1. Create a new group called `robotics_firmware`.
`sudo groupadd robotics_firmware`
2. Create a new user named `developer1`.
`sudo useradd -m developer1`
3. Set a password for `developer1`.
`sudo passwd developer1`
4. Add `developer1` to the `robotics_firmware` group.
`sudo usermod -aG robotics_firmware developer1`
5. Verify that `developer1` is now in the group.
`id developer1`
6. Create a project directory in `/opt` (a common place for optional software).
`sudo mkdir /opt/firmware_project`
7. Change the ownership of the directory so it is owned by the root user and the `robotics_firmware` group.
`sudo chown root:robotics_firmware /opt/firmware_project`
8. Set permissions so only the owner (root) and group members can read, write, and access the directory. No one else gets access (770).
`sudo chmod 770 /opt/firmware_project`
9. Verify the ownership and permissions.
`ls -ld /opt/firmware_project`
10. Finally, clean up by deleting the user and group.
`sudo userdel -r developer1`
`sudo groupdel robotics_firmware`

The Idea Behind the Assignment

Why is this crucial for Intelligent Systems? **The Principle of Least Privilege.**

- **IoT Device Security:** An IoT device might run several services. By creating a separate user and group for each service, you ensure that if one service is compromised, the attacker's access is limited. Each part of the system only has the permissions it needs.
 - **Android App Sandboxing:** Android is built on Linux. Every Android app you install is assigned a unique UID and runs as its own user. This "sandboxing" is why a calculator app cannot read your contacts—it's a different user with no permissions to access that data.
 - **Collaborative Development:** When a team works on a shared device (like a Raspberry Pi), groups ensure that team members can access project files while keeping the core OS files safe from accidental changes.
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Assignment 3: User & Group Administration (12%)

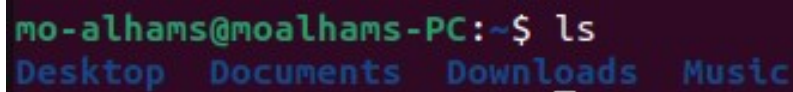
Objective: To practice creating users and groups to manage file access securely.

Tasks:

1. Create a new group named `project_team`.
 2. Create two new users: `manager` and `developer`. Be sure to create their home directories.
 3. Set simple passwords for both users.
 4. Add both `manager` and `developer` to the `project_team` group.
 5. Create a directory: `sudo mkdir /srv/project_alpha`. (`/srv` is for site-specific data).
 6. Change the directory's group ownership to `project_team`.
 7. Set the directory's permissions so that the owner (`root`) and members of `project_team` can read, write, and access it, but other users have zero access (`rw-rw-r--` or `770`).
 8. **Verification:** Run the command `ls -ld /srv/project_alpha` and include its output (as text or screenshot) in your report to prove you have set ownership and permissions correctly.
 9. **Report:** Write a brief paragraph (3-5 sentences) explaining the main benefit of using a group to manage access to `/srv/project_alpha` instead of assigning permissions to `manager` and `developer` individually.
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Assignment Submission Instructions

- **File Name:** Name your submission file exactly as follows:
studentnum-studentname-Assignment3.pdf
 - *Example:* 13012070xx-MohammedEmadAlhams-Assignment3.pdf
- **Terminal Prompt:** Your name must appear in the terminal prompt for your screenshots, as shown in the example below.



```
mo-alhams@moalhams-PC:~$ ls
Desktop  Documents  Downloads  Music
```

- **File Format:** Your submission must be a **single PDF file** containing:
 - The ~~screenshot for the verification step.~~
 - A description of each command used and its functionality.
 - Your report paragraph.
 - **Academic Integrity:**
 - **Warning:** Any two similar submissions will be considered an attempt to cheat and will result in a grade of **ZERO**.
 - **Warning:** Any AI-generated solutions or reports will also result in a grade of **ZERO**.
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Next Steps & Resources

Practice This Week:

- Get comfortable creating and deleting test users. Remember to use -r with userdel to keep your system clean.
- Explore the /etc/passwd and /etc/group files with cat or less. Can you find the users and groups you created?

Next Week: Lecture 5 - Process Management in Linux

- Now that we have users, we'll look at what happens when they run programs. We will learn how to view, manage, and prioritize the processes running on our system.

Resources:

- Ubuntu Server Guide - User Management: <https://ubuntu.com/server/docs/user-management>
- Read the manual pages: man useradd, man usermod, man groupadd
- Course Materials on Moodle